# 3.6 Graphing Linear Equations in Slope-Intercept Form

Name _____ Period ____ Date _____

Algebra I 3.6 · Student sheet

**I can…**

- find the slope of a line from a graph, two points, or a table.
- identify the slope and y-intercept and graph y = mx + b.
- interpret the slope and y-intercept in real-life situations.

**Vocabulary** · fill in the missing word

**Slope** — m = _____/run = $\frac{y_2 - y_1}{x_2 - x_1}$

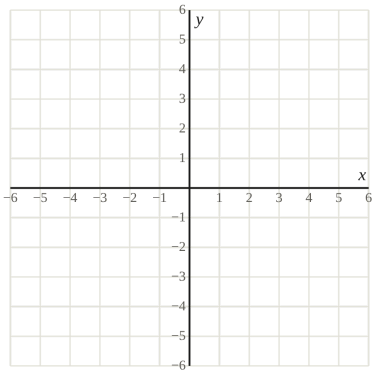
**Slope-intercept form** — y = mx + b: slope m, _____ b

**Kinds of slope** — _____ (rises), negative (falls), zero (horizontal), undefined (vertical)

## Part 1 · Walk the line

Graph all three lines: start at the y-intercept, then use the slope to walk to more points.

y = 2x + 1 y = −x + 4 y = ½x − 2




| Line | Slope | Positive or negative? |
|---|---|---|
| y = 2x + 1 | | |
| y = −x + 4 | | |
| y = ½x − 2 | | |

## Part 2 · Notes: fill in the blanks

Slope m = _____ / _____ = $(y_2 - y_1)/($_____$)$

In y = mx + b, m is the _____ and b is the _____.

A horizontal line has slope _____; a vertical line has _____ slope.

## Part 3 · Examples (work them with the class)

### Example 1

Find the slope of the line through $(-2, 5)$ and $(4, -1)$.

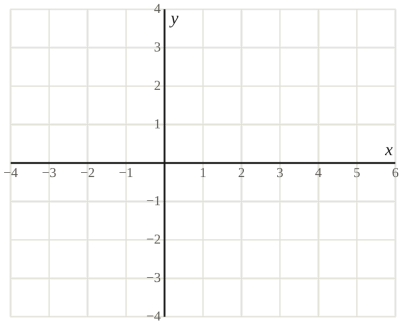
### Example 2

Find the slope and y-intercept.

a. $y = -3x + 4$ b. $2x + y = 5$

### Example 3

Graph $y = (\frac{2}{3})x - 1$. Find the x-intercept.




### Example 4

The cost of a party is $y = 12x + 50$, where $x$ is the number of guests. What do the slope and y-intercept mean?

## Part 4 · Partner practice

### Partner 1

Find the slope from the table:  x: 0, 1, 2;  y: 7, 4, 1

### Partner 2

Find the slope and y-intercept of $y = 6 - x$.